Yashas Nagesh Gowda

Machine Learning Engineer

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Summary

Machine Learning Engineer with 5+ years of experience taking **machine learning and deep learning** models from ideas to production across cloud and on-premises environments. Hands-on with **Python, R, TensorFlow, PyTorch, Scikit-learn, SQL, and Spark**, with real-world work in **NLP, computer vision, recommendation systems, fraud detection, and Generative AI**, including LLM fine-tuning and semantic search with vector databases. Comfortable owning the full ML lifecycle: building data pipelines with **Airflow and Kafka**, deploying models on **AWS SageMaker, GCP Vertex AI, and Azure ML**, and keeping them reliable in production with **Docker, Kubernetes, MLflow, and CI/CD** as part of strong **MLOps** practices. Focused on building models that stay accurate after launch, scale to millions of users, and help teams ship faster.

Core Skills

Programming & Languages: Python, R, Java, Scala, C++, Bash

Machine Learning & AI: TensorFlow, PyTorch, Keras, Scikit-learn, XGBoost, LightGBM, CatBoost, AutoML

Deep Learning: CNNs, RNNs, LSTMs, GRUs, Transformers, GANs, Vision Transformers (ViT)

NLP: Hugging Face, OpenAI APIs, BERT, GPT, T5, spaCy, NLTK, Word2Vec, Doc2Vec, FastText

Computer Vision: OpenCV, YOLO, ResNet, EfficientNet, OCR, Mask R-CNN

Data & Big Data: SQL, NoSQL (MongoDB, Cassandra, DynamoDB), Pandas, NumPy, PySpark, Hive, Hadoop, Presto, Dask

Cloud & MLOps: AWS (SageMaker, S3, Lambda, EC2, EMR, Redshift, Glue, Kinesis), GCP (Vertex AI, BigQuery, Dataflow, Pub/Sub), Azure ML, MLflow, Kubeflow, Airflow, Kafka, Docker, Kubernetes, Jenkins, Git/GitHub Actions

Deployment & Monitoring: FastAPI, Flask, Streamlit, REST APIs, GraphQL, Prometheus, Grafana, Seldon Core

Visualization & BI: Tableau, Power BI, Matplotlib, Seaborn, Plotly

Other Tools: Jupyter Notebook, Colab, VS Code, PyCharm, Unix/Linux, Terraform, Anaconda

Professional Experience

Machine Learning Engineer — OpenAI — Dec 2024 – Current

- Developed and fine-tuned NLP models with Hugging Face Transformers and OpenAI APIs, improving summarization and classification accuracy by 32%.
- Built ML pipelines with Airflow and MLflow, automating ingestion, training, validation, and deployment, reducing manual effort by 45%.
- Deployed real-time inference services using FastAPI, Docker, and Kubernetes, achieving 99.9% uptime and scaling to 1M+ daily API calls.
- Integrated vector databases (FAISS, Pinecone, Weaviate) for semantic search, improving retrieval speed by 40%.
- Applied hyperparameter tuning with Optuna and Ray Tune, boosting F1 score by 18%.
- Built monitoring dashboards in Prometheus and Grafana to detect drift and latency issues, reducing downtime by 60%.
- Created automated retraining workflows triggered by data drift detection in AWS SageMaker, ensuring models remained production ready.
- Worked on Generative AI chatbots with transformer-based embeddings, improving contextual accuracy by 25%.

Machine Learning Engineer — Amazon — Jun 2019 – Jul 2023

- Designed recommendation systems using collaborative filtering and deep learning, increasing CTR by 27% and conversions by 18%.
- Processed 10TB+ datasets with PySpark, Hive, and AWS EMR, cutting ETL runtime by 35% and reducing costs by 20%.
- Built fraud detection models using Isolation Forests, Autoencoders, and XGBoost, reducing false positives by 22%.
- Created real-time ingestion pipelines with Kafka and AWS Glue, reducing latency by 40% and improving reliability.
- Automated model retraining with AWS SageMaker, CI/CD pipelines, and Docker, improving accuracy by 20%.
- Implemented MLOps workflows using Jenkins, Kubernetes, and GitHub Actions, cutting deployment cycle by 30%.
- Developed APIs for ML models with Flask and REST, integrating into Amazon e-commerce platforms serving millions of customers.
- Improved feature engineering using PCA, t-SNE, and embedding models, cutting training time by 15%.
- Optimized computer vision models with OpenCV, YOLO, and CNNs for product classification, achieving 95%+ accuracy.
- Designed dashboard visualizations with Tableau and Power BI for model insights, improving stakeholder decision-making speed by 35%.

Education

Masters in computer science — Florida State University, Tallahassee

Certifications

Microsoft Certified – Azure Data Scientist Associate

Microsoft Certified – Power BI Data Analyst Associate

Database and SQL for Data Science with Python